

JavaScript mid-term exam (2019-20-2)

Due No Due Date **Points** None

Information

- You have **150 minutes** to complete the tasks.
- You have to upload the finished files to Canvas. **At 7:30pm (19:30) the system will close, after that you will not be able to upload anything.**
- The tasks do not depend on one another, you can solve them in **any order**.
- Each task is worth the same amount of points.
- Create **separate files** for each of the tasks:

Task 1: ,

Task 2: ,

Task 3: ,

Task 4: ,

- Create a file containing the following statement (without the <> signs):

```
<Name>
<Neptun ID>
<Subject & Assignment name>
<Submission date>
This solution was submitted and prepared by <Name, Neptun ID> for the JavaScript Mid-term Exam of the <Course name> course.
I declare that this solution is my own work. I have not copied or used third party solutions. I have not passed my solution to my classmates, neither made it public. Students' regulation of Eötvös Loránd University (ELTE Regulations Vol. II. 74/C. § ) states that as long as a student presents another student's work - or at least the significant part of it - as his/her own performance, it will count as a disciplinary fault. The most serious consequence of a disciplinary fault can be dismissal of the student from the University.
```

- In the file, under the statement (separated by an empty line) create the following list. Fill each ☐ with an ☐ if you solved the corresponding sub-task (even if only partially)!

```
- Task 1
+ ☐ Sub-task 1
+ ☐ Sub-task 2
+ ☐ Sub-task 3
- Task 2
+ ☐ Sub-task 1
+ ☐ Sub-task 2
+ ☐ Sub-task 3
+ ☐ Sub-task 4
- Task 3
+ ☐ Sub-task 1
+ ☐ Sub-task 2
+ ☐ Sub-task 3
- Task 4
```

- + [] Sub-task 1
- + [] Sub-task 2
- + [] Sub-task 3

Without the proper `README.md` file your submission will be declined!

Task 1

Lore

On today's press conference the National Health Organization announced that they will be closing down Hungarian torrent clients - the goal is to slow the spread of the coronavirus through online platforms. Of course you as an educated programmer know that torrents can't infect PC-s, because nobody would download anything illegal ever. For this reason you take on the task of writing your own torrent client, which will be much better than all the others.

Task

We are given a table that stores our downloads. Every download has a name, size (in NotfakeBits, NfB) and progress (percentage).

```
<table>
  <thead>
    <tr>
      <th>Name</th>
      <th>Size</th>
      <th>Progress</th>
    </tr>
  </thead>
  <tbody>
    <tr>
      <td>Sonic the Hedgehog eng 720p</td>
      <td>100NfB</td>
      <td>25%</td>
    </tr>
    <tr>
      <td>Frozen 2 eng 1080p</td>
      <td>200NfB</td>
      <td>0%</td>
    </tr>
    <tr>
      <td>The Sims 4 all dlc included</td>
      <td>1500NfB</td>
      <td>10%</td>
    </tr>
  </tbody>
</table>
```

1. Create an input field, where we can set the download speed (in NfB/s), and an accompanying "Download" button.
2. Pressing the "Download" button should cause the program to print the total size and the downloaded size (NfB) of each torrent to the console.
3. Pressing the "Download" button should cause the program to increase the progress of every download with the given amount. The progress percentage must be updated accordingly. If a download reaches 100%, it should not increase anymore.

While completing the task, do not rely on the fact that you know the amount of torrents (rows). The script should properly run with an arbitrarily large table!

Download speed:

Name	Size	Progress
Sonic the Hedgehog eng 720p	100NfB	100%
Frozen 2 eng 1080p	200NfB	56.000000000000001%
The Sims 4 all dlc included	1500NfB	17.466666666666665%

Task 2

Lore

While preparing for the zombie apocalypse coming in April, one of Hungary's public institutions started gathering strategic resources. The incoming cargo is placed in a secret warehouse with the utmost care in the 11th district. Based on a brilliant idea the data is stored in HTML instead of a database, so now only a JavaScript program is missing to manage the data. The creation of this script was delegated to you.

Task

We are given a list that stores incoming shipments. Each shipment has an arrival date, a list of the products it contains (an embedded list) and an assigned shelf on which it is stored.

```
<style>
  .product { background: lime; }
  .arrival { font-weight: bold; }
  .shelf { font-style: italic; }
</style>
<div id="data">
  <ul>
    <li class="shipment">
      <div>Arrival: 03.31. 2020.</div>
      <div>
        <ul>
          <li>Toilet paper</li>
          <li>Hand sanitizer</li>
          <li>Medical mask</li>
        </ul>
      </div>
      <div>Shelf: 2187</div>
    </li>
  </ul>
</div>
```

```

</li>
<li class="shipment">
  <div>Arrival: 03.30. 2020.</div>
  <div>
    <ul>
      <li>PureStar soft drink</li>
      <li>Antacid</li>
    </ul>
  </div>
  <div>Shelf: 42</div>
</li>
<li class="shipment">
  <div>Arrival: 03.29. 2020.</div>
  <div>
    <ul>
      <li>Haribo gummy bears</li>
      <li>Smirnoff juice (100% fruit)</li>
    </ul>
  </div>
  <div>Shelf: 501</div>
</li>
</ul>
</div>

```

1. If we click a product (eg.: *Hand sanitizer*) the given list item's background should be changed to green (use the `.product` class).
2. If we click arrival (eg.: *Arrival: 03.29. 2020.*) the whole shipment's font should be changed to bold (use the `.arrival` class).
3. If we click shelf (eg.: *Shelf: 501*) the whole shipment's font should be changed to italic (use the `.shelf` class).
4. If we click any of the above again then its corresponding formatting should be removed (green, bold or italic).

While completing the task, do not rely on the fact that you know the amount of shipments (list items). The script should properly run with an arbitrarily large list. Do not use loops or iterators to append events (use delegation).

- **Arrival: 03.31. 2020.**
 - Toilet paper
 - Hand sanitizer
 - Medical mask**Shelf: 2187**
- *Arrival: 03.30. 2020.*
 - PureStar soft drink
 - Antacid*Shelf: 42*
- *Arrival: 03.29. 2020.*
 - Haribo gummy bears
 - Smirnoff juice (100% fruit)*Shelf: 501*

Task 3

Lore

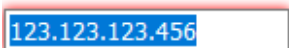
The NATO alerted all its members that May won't be easy, as the Klingon Defence Force will attack Earth. With some luck international informants stole the invasion plans with a cunning move, so we have a chance to prepare. Their wily generals want to surprise us, and came up with a plan to change internet protocols so our browsers will forward users to corrupted Klingon pages. As the best hacker in the EU, you've been tasked to prevent this behaviour.

Task

You are given an input field that is used to input an IP address.

```
<style>
  .incorrect { box-shadow: 0 0 5px red; }
  .correct   { box-shadow: 0 0 5px lime; }
</style>
<input>
```

1. As NATO expects outstanding precision from all its employees, you must disable the option to delete text from the input field using the Backspace or Del buttons (these buttons should not do anything).
2. Make sure that the input field only accepts characters required for an IP address (integers and dots - pressing other characters on the keyboard should have no effect).
3. If the content of the input field changes, check whether it's a valid IP address (the format is: `[0-255].[0-255].[0-255].[0-255]`). If the formatting is correct, apply the `.correct` class to the input field, but if the formatting is incorrect, apply the `.incorrect` class.



Pressed button: Backspace

Note: you don't have to show the pressed button below the input as in the example, it is there for demonstration purposes.

Task 4

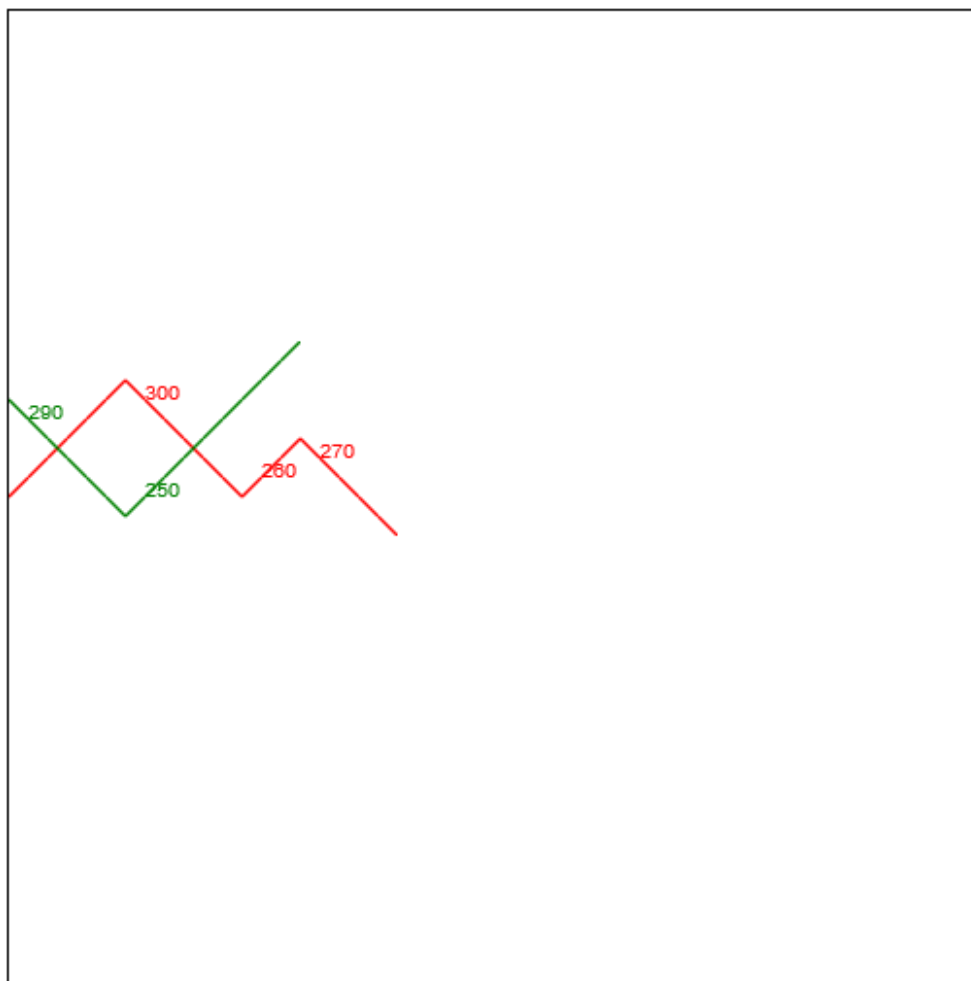
Lore

After the virus, the zombies and the attack of the Klingons, July brings the promise of a more peaceful time, and we have time to reflect on recent events. The Department of Human Resources would like to summarise the changes in the two main economic indicators using graphs: the price of BigMac and Spotify Premium (in NotfakeCurrency, NfC). Because you are the most promising artist, you've been asked to paint the data onto your canvas.

Task

Create a page where you can illustrate the changes of these values with a graph. We know that the base price of BigMac is 250NfC and Spotify premium subscription is 300 NfC.

1. Create two buttons that change the price of BigMac. If you press the up button the price of the BigMac jumps to the next day and grows by 10, if you press the down button it decreases by 10. The colour of BigMac data is red.
2. Create two buttons that change the price of Spotify. If you press the up button the price of the Spotify jumps to the next day and grows by 10, if you press the down button it decreases by 10. The colour of Spotify data is green.
3. If either of the curves' slope changes its sign (goes from negative to positive or positive to negative), write the current value of the given product (BigMac or Spotify) on the graph.



BigMac+	Spotify+
BigMac-	Spotify-

Mid-term JS test

Criteria	Ratings		Pts
Task 1	7.5 pts Full Marks	0.0 pts No Marks	7.5 pts
Task 2	7.5 pts Full Marks	0.0 pts No Marks	7.5 pts
Task 3	7.5 pts Full Marks	0.0 pts No Marks	7.5 pts
Task 4	7.5 pts Full Marks	0.0 pts No Marks	7.5 pts
			Total Points: 30.0